

Question:
At birth, the least developed part of the brain is the

- Answer Choices:
- A. visual system
 - B. thalamus
 - C. frontal lobe
 - D. brain stem
 - E. parietal lobe
 - F. cerebellum
 - G. cortex
 - H. basal ganglia
 - I. hypothalamus
 - J. limbic system

Reasoning + Answer:

<think>
I can provide a detailed step-by-step chain-of-thought, but here is a concise, high-level explanation:

At birth, the human brain is extremely immature in many functions compared to adult anatomy. The structures that rely most heavily on development after birth and are the last to mature are: the frontal lobes (executive and social/linguistic control), the visual system (color vision, complex visual recognition), and particularly the brain stem reticular formation (wake-sleep regulation and many homeostatic reflexes). In contrast, the thalamus is already relatively mature at birth and primarily performs early relay roles; the cerebellum is mature for coordinated voluntary movement; the basal ganglia contribute to posture and movement control from a young age; and the hypothalamus (especially its neuroendocrine functions) is relatively well-developed. While the cortex, as a whole, appears gray matter-rich at birth (with substantial development afterward), the question specifically points to the least developed part in infancy, which is most clearly the brain stem.

Correct answer: D. brain stem

</think>

<answer>

D. brain stem

</answer>

